

BRD & FRD: Buy Now, Pay Later (BNPL) Risk Engine

Product overview	
 Target date	Aug 6, 2026
 Document status	Completed
 Team members	SHARATH C
Quick links	
 Designs	https://www.figma.com/board/ckseOqTHeePztuw2dF7IVU/BNPL-Checkout-Process-Flow?node-id=0-1&t=Wlz3MgmgzthipKaM-1

Objective

The business is experiencing an unacceptable rate of first-payment defaults within the Buy Now, Pay Later (BNPL) checkout option, which is negatively impacting monthly recurring revenue (MRR). The objective of this project is to implement a dynamic, machine-learning-driven risk scoring engine directly into the checkout flow. By predicting a user's probability of default in real-time using pre-purchase financial indicators, the business can intelligently segment users and enforce dynamic down-payment requirements, thereby mitigating financial risk while preserving conversion rates for low-risk users.

-  • Deploy a predictive classification model (Random Forest) to accurately assess the default probability of users opting for BNPL.
- Achieve a minimum 80% recall rate for identifying high-risk defaults prior to transaction approval.
- Reduce overall first-payment default rates by 15% in Q3 by dynamically adjusting down-payment requirements based on real-time risk segmentation.

Project Scope

In-Scope

- Extraction and analysis of pre-purchase user data (e.g., Credit Score, Monthly Income, Debt-to-Income Ratio).
- Development and deployment of a predictive classification model.
- Creation of an interactive Power BI dashboard for ongoing risk monitoring.
- Integration of risk tiers (Low, Medium, High) into the checkout decision engine.

Out-of-Scope

- Analysis of post-purchase repayment behavior (e.g., repayment delay days, missed payments).
- Modifications to the core credit card payment gateway



Requirements

Req ID	Description	Priority
BR-01	The system must capture real-time financial indicators (Credit Score, Income, Debt-to-Income) at the moment a user selects the BNPL option.	High
BR-02	The predictive model must return a probability score and classify the user into Low, Medium, or High risk tiers within 2 seconds.	High
BR-03	Users classified as "High Risk" must be denied the BNPL option and redirected to standard credit card checkout.	High
BR-04	Users classified as "Medium Risk" must be prompted to pay a mandatory 25%	Medium

	upfront down-payment to proceed with BNPL.	
BR-05	A Power BI monitoring dashboard must be updated daily to track the distribution of risk segments and model performance.	Medium
FR-01	The system shall extract <code>credit_score</code> , <code>monthly_income</code> , and <code>debt_to_income_ratio</code> from the user profile database via API at the exact moment the BNPL checkout button is clicked.	High
FR-02	The Python Random Forest model must process the extracted variables and return a <code>default_probability</code> float value between 0.0 and 1.0 within a maximum latency of 2 seconds .	High
FR-03	The system's decision engine must classify the default probability as follows: • Score ≥ 0.60 → High Risk • Score 0.40 to 0.59 → Medium Risk • Score < 0.40 → Low Risk	High
FR-04	For "Medium Risk" users, the checkout UI must dynamically render a	Medium

	payment gateway prompt requiring a 25% down-payment before finalizing the BNPL contract.	
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Key Success Metrics

- **Recall Rate (Class 1 Defaults):** Target > 85%.
- **BNPL Default Rate:** Target reduction of 15%.
- **Checkout Conversion Rate (Low Risk):** Must remain stable or increase.